

MOHAMAD KHALIF

Geodetic Engineer | Geospatial Analyst | Remote Sensing Enthusiast

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SUMMARY

Geodetic Engineering graduate from Institut Teknologi Nasional Malang with cross-functional experience combining technical geospatial expertise and data analysis precision. Began my career as an Assistant Surveyor Intern at PT Virama Karya, conducting topographic surveys, staking out, and cut-and-fill volume calculations to support toll road construction projects. This experience was further strengthened through my role as a Warehouse Data Analyst at PT. Thousand Island Services, where I managed inventory analysis, stock opname, and systematic operational reporting, as well as my experience as a Content Moderator at TELUS Digital, which sharpened my discipline in working within strict standards and procedures. Proficient in GIS (ArcGIS, QGIS), remote sensing (Google Earth Engine, ENVI), and Python programming for spatial analysis, orthophoto processing, WebGIS development, and the application of AI/ML in geospatial analysis. The combination of field survey experience, data analysis, and geospatial technology proficiency makes me ready to contribute to projects requiring both technical precision and analytical capability.

CORE SKILLS

- GIS Software — ArcGIS, QGIS
- Remote Sensing — Google Earth Engine, ENVI
- CAD & Photogrammetry — AutoCAD Civil 3D, Agisoft Metashape
- Geospatial Web & Databases— WebGIS, GeoServer, PostgreSQL/PostGIS
- Programming — Python, JavaScript, MATLAB
- Survey Equipment— GNSS, Total Station, Waterpass (Level), Theodolite
- Productivity Tools — Microsoft Excel, Microsoft Word, Microsoft Power point

PROFESSIONAL EXPERIENCE

Content Moderator

Jun 2025 – Sep 2026

TELUS Digital

- Moderated user-generated content in accordance with platform guidelines
- Managed and organized content for review and reporting purposes

Ensured compliance with platform terms and conditions

Warehouse Data Analyst

Des 2025 – Mei 2026

PT. Thousand Island Services

- Conducted inventory analysis and stock control
- Performed stock opname (physical inventory counts) and managed spare parts distribution
- Prepared warehouse reports and analyzed operational data

Assistant Surveyor Intern

Jul 2024 – Sep 2024

PT. Virama Karya

- Conducted topographic surveys and staking out
- Performed cut-and-fill (excavation and embankment) volume calculations
- Supported toll road construction survey activities

EDUCATION

B.Eng., Geodetic Engineering

2021 – 2025

National Institute of Technology Malang

Portfolio

About Me

Experiences

Project

Certificate



Mohamad Khalif

Portfolio

GEODETIC
ENGINEER

Introduction

I'm a Geodetic Engineering fresh graduate, a little mix between a researcher and a field practitioner. I'm quite adaptive, can work independently but also enjoy teamwork. I love problem-solving, critical thinking, and paying attention to details is really important for me. I'm also the type who's open to feedback and learns new things quickly whether it's tools, workflow, or project challenges. I have experience in several academic and field projects. I'm very interested in terrestrial surveying, remote sensing, and cloud-based data platforms like Google Earth Engine to create interactive web-based maps. Besides that, I'm also interested in AI/ML, especially in geospatial analysis. During college, I've been involved in various projects such as terrestrial surveys, hydrography, aerial mapping, and of course remote sensing as my favorite field. During my internship in the Probolinggo Toll Road project, I handled staking out points and land volume calculations for cut-and-fill analysis based on field

coordinates. Throughout my academic journey, I'm quite fluent in using geospatial software such as ArcMap, QGIS, and ENVI. Coding is also an important part of my work: Python, PostgreSQL, Java (Google Earth Engine), and MATLAB are my daily tools. I have used TensorFlow & Keras to build vegetation classification models from high-resolution satellite imagery, and in my thesis, I successfully analyzed and created a multitemporal aerosol distribution map of Jakarta (2021-2024) using Sentinel and MATLAB. My friends usually call me "the vibe coder" because I love exploring new things and finding ways to blend geospatial science, remote sensing, and AI/ML into impactful solutions.



PT. Virama Karya (Probowangi Toll Road Package 1)

During this internship, I served as a surveyor assistant, responsible for staking out the Top Subgrade, Lean Concrete, and Rigid Pavement to ensure precise alignment with the planned coordinates and elevations. In my final internship report, I performed land volume calculations using long and cross-section maps to identify cut and fill areas based on the marking coordinates.



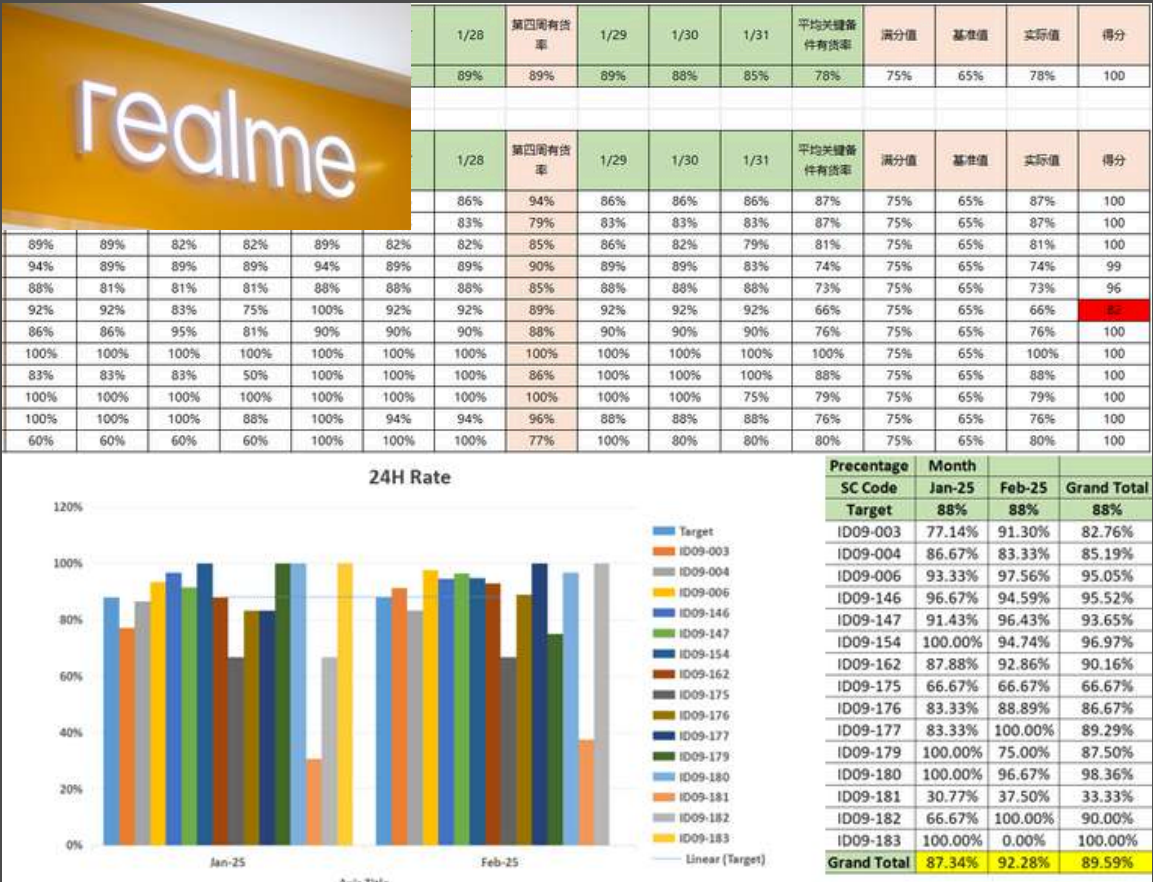
PTSL (Pendaftaran Tanah Sistematis Lengkap) Lamongan, Kec. Solokuro

In this experience, I was given the opportunity to participate in the PTSL program with a project to map several village boundaries with local residents and determine and supervise the installation of ICP (Independent Control Point) markers for photography from a UAV.



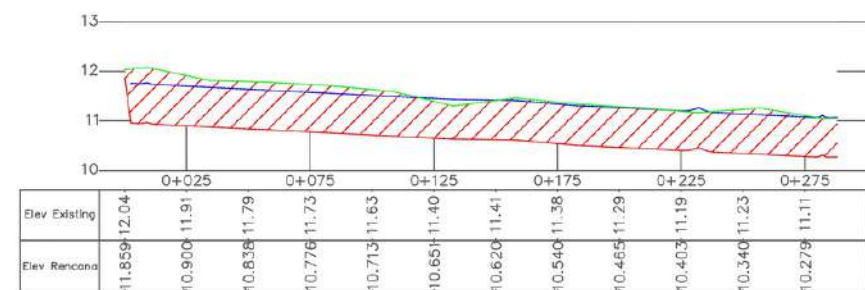
PT. PT Thousand Island Services

I am experienced in handing end-to-end warehouse administration processes including stock opname, returnmanagement, claim processing, work order monitoring, and unit replacement coordination. I also manage inter-agent spare part transfers to optimize stock distribution and minimize shortages.



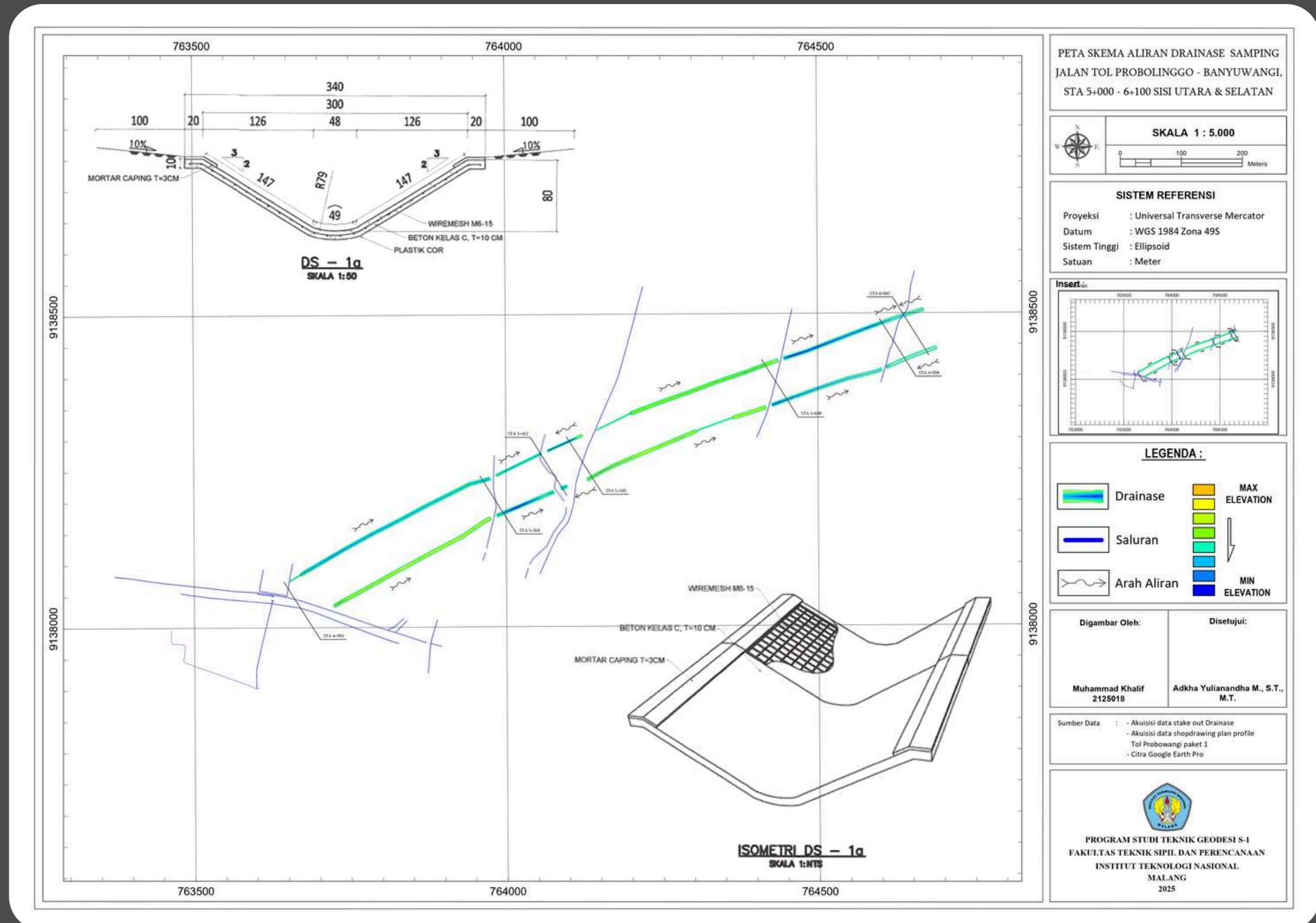
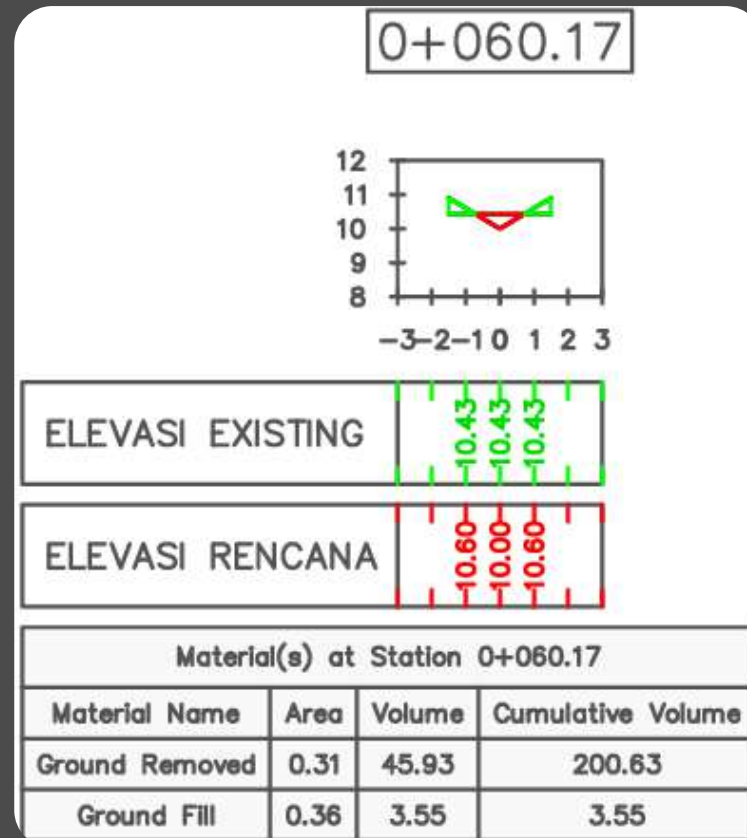
Cut & Fill Project

Station	Cut Area (Sq.m.)	Cut Volume (Cu.m.)	Reusable Volume (Cu.m.)	Fill Area (Sq.m.)	Fill Volume (Cu.m.)	Cum. Cut Vol. (Cu.m.)	Cum. Reusable Vol. (Cu.m.)	Cum. Fill Vol. (Cu.m.)	Cum. Net Vol. (Cu.m.)
0+000.066	1.58	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
0+025.185	5.31	86.54	86.54	0.00	0.00	86.54	86.54	0.00	86.54
0+050.459	5.74	139.75	139.75	0.00	0.00	226.29	226.29	0.00	226.29
0+075.521	6.30	150.93	150.93	0.00	0.00	377.22	377.22	0.00	377.22
0+100.979	6.03	156.95	156.95	0.00	0.00	534.17	534.17	0.00	534.17
0+126.201	5.68	147.67	147.67	0.00	0.00	681.84	681.84	0.00	681.84
0+151.428	5.81	144.97	144.97	0.00	0.00	826.81	826.81	0.00	826.81
0+176.657	5.47	142.40	142.40	0.00	0.00	969.21	969.21	0.00	969.21
0+201.874	5.34	136.32	136.32	0.00	0.00	1105.53	1105.53	0.00	1105.53
0+227.099	5.29	134.00	134.00	0.00	0.00	1239.53	1239.53	0.00	1239.53
0+252.103	5.13	130.27	130.27	0.00	0.00	1369.80	1369.80	0.00	1369.80
0+277.549	5.13	130.58	130.58	0.00	0.00	1500.38	1500.38	0.00	1500.38
0+302.767	4.88	126.20	126.20	0.00	0.00	1626.58	1626.58	0.00	1626.58

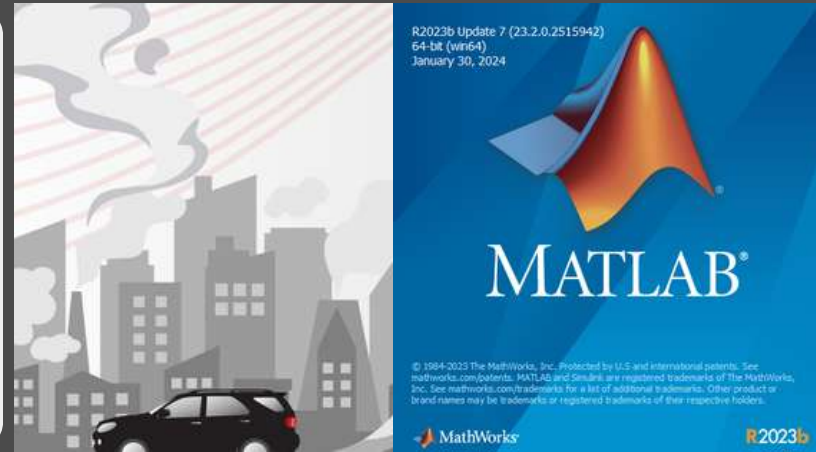
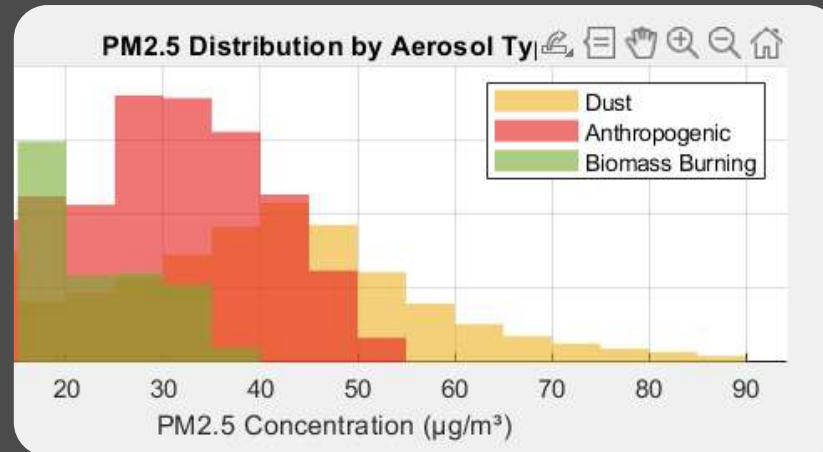


Terrestrial Survey Strategy for Determining Cut and Fill Volumes of Side Drainage in the Planning

This study aims to design a terrestrial survey strategy to support the calculation of cut and fill volumes for side drainage channels in the Probawang Toll Road construction project, specifically at STA 5+000 - 6+000. The approach utilizes existing elevation data and planned drainage design data to analyze elevation differences between the current ground surface and the proposed drainage channel design. This calculation is essential to ensure optimal earthwork balance, improve drainage flow efficiency, and support a sustainable and cost-effective construction process.

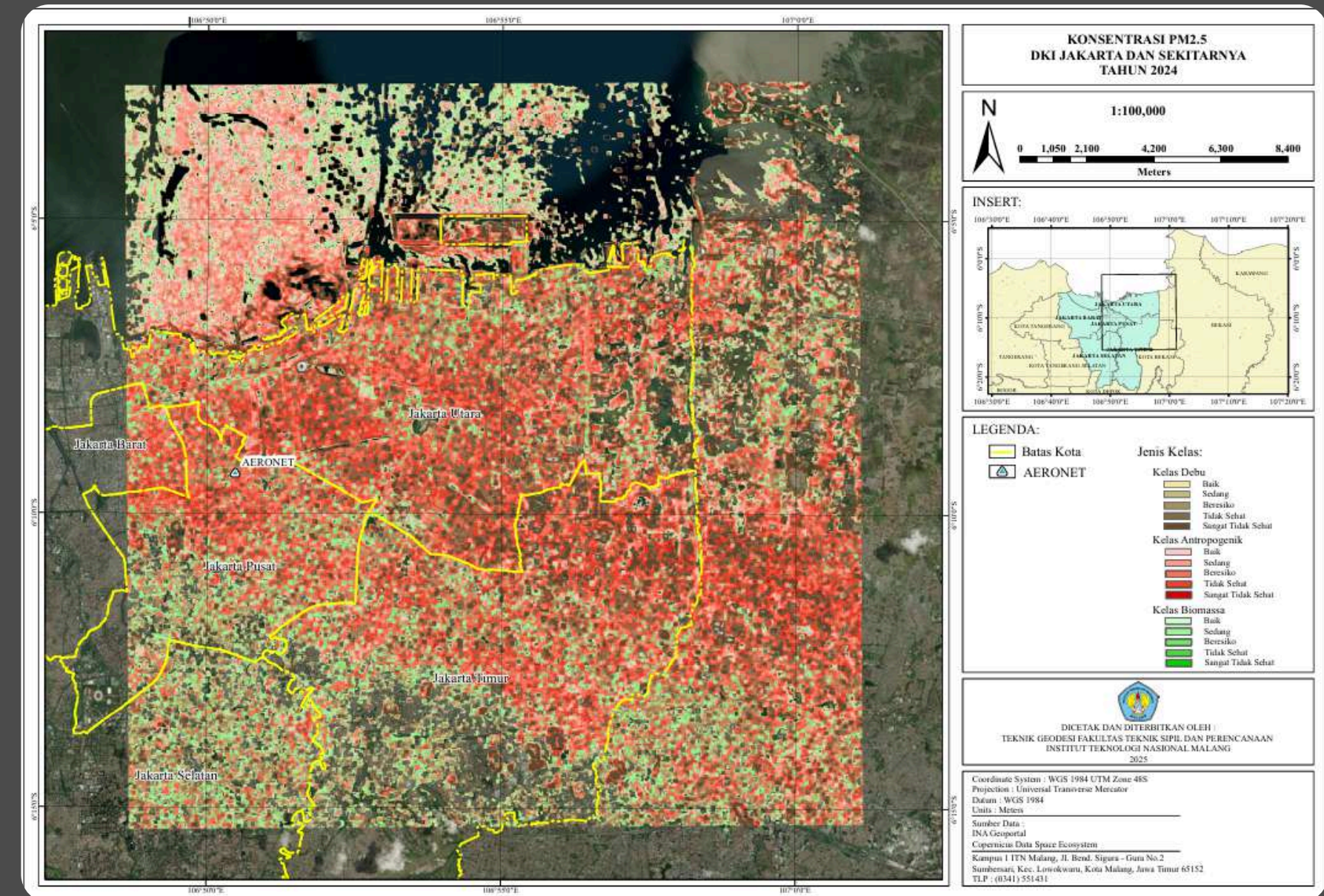


Recent Studies



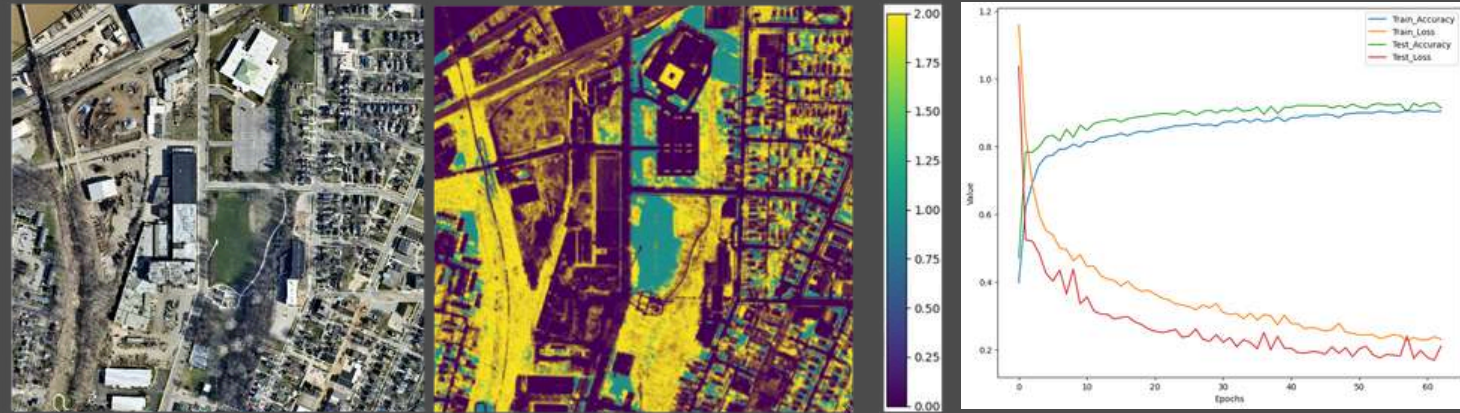
Analysis of PM2.5 Main Component Concentration Using Multitemporal Satellite Imagery and AERONET Data

This research focused on the Jakarta Metropolitan Area (DKI Jakarta and surroundings) and analyzed the concentration of PM2.5 using multi-temporal satellite data from Sentinel-2A and Landsat-8 OLI. The study applied several atmospheric analysis methods, including the Dispersion Coefficient (DC), Ångström Exponent (AE), and Normalized Gradient Aerosol Index (NGAI) to estimate aerosol distribution and identify dominant particulate components. Validation was conducted using Aerosol Optical Depth (AOD) data from the AERONET BMKG Jakarta site to ensure model accuracy and reliability.



[Click here for More information about this project](#)

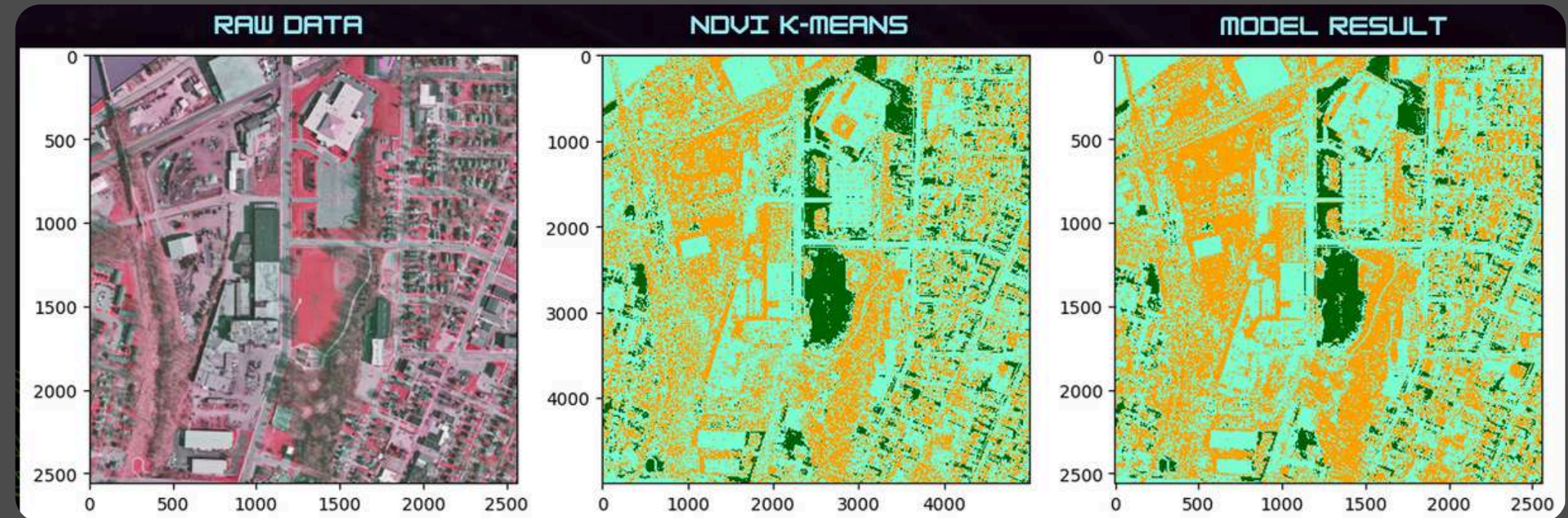
DeepLearn (AI)



Built an NDVI-based K-Means classification model for vegetation analysis using TensorFlow and Keras.

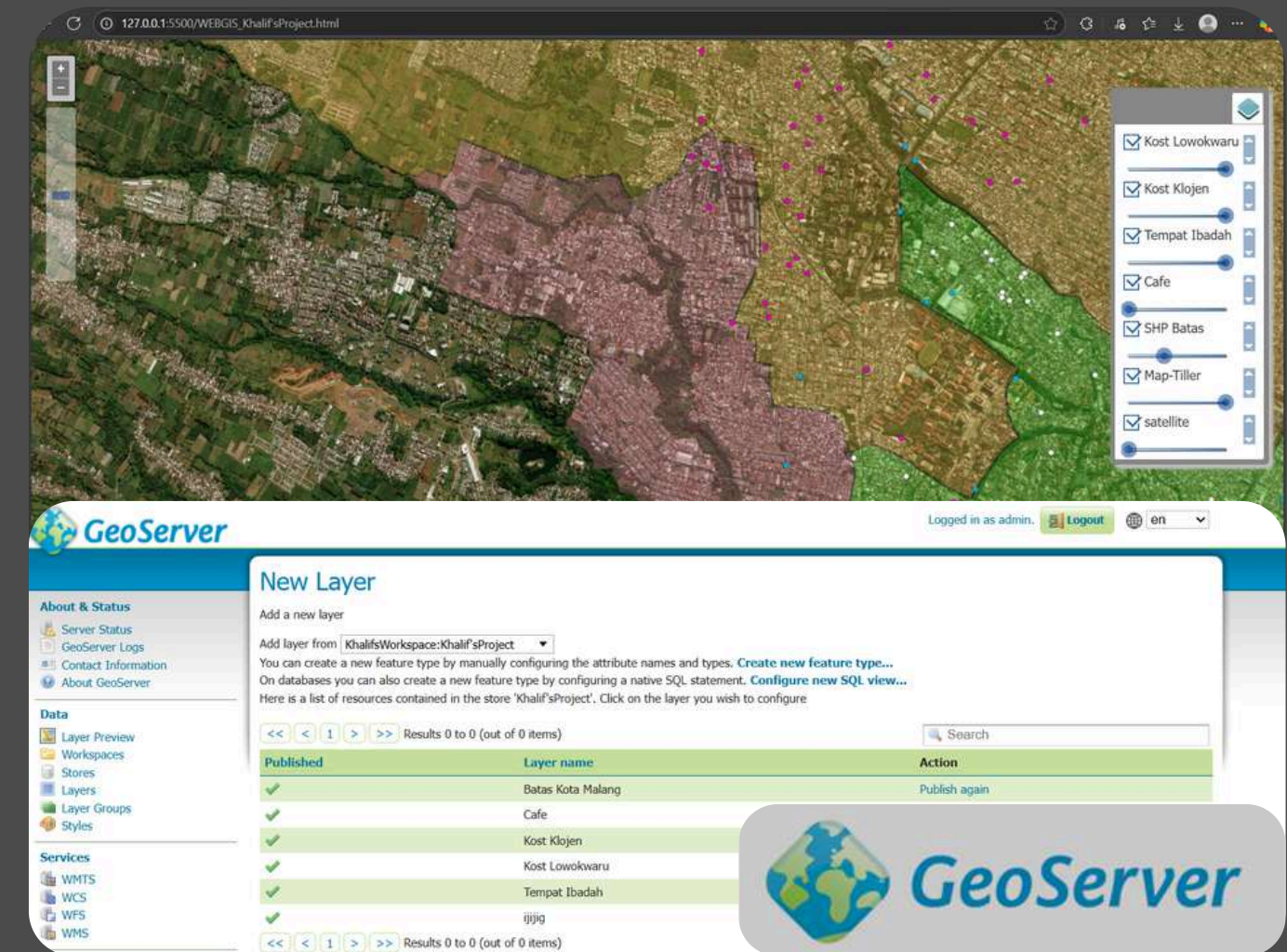
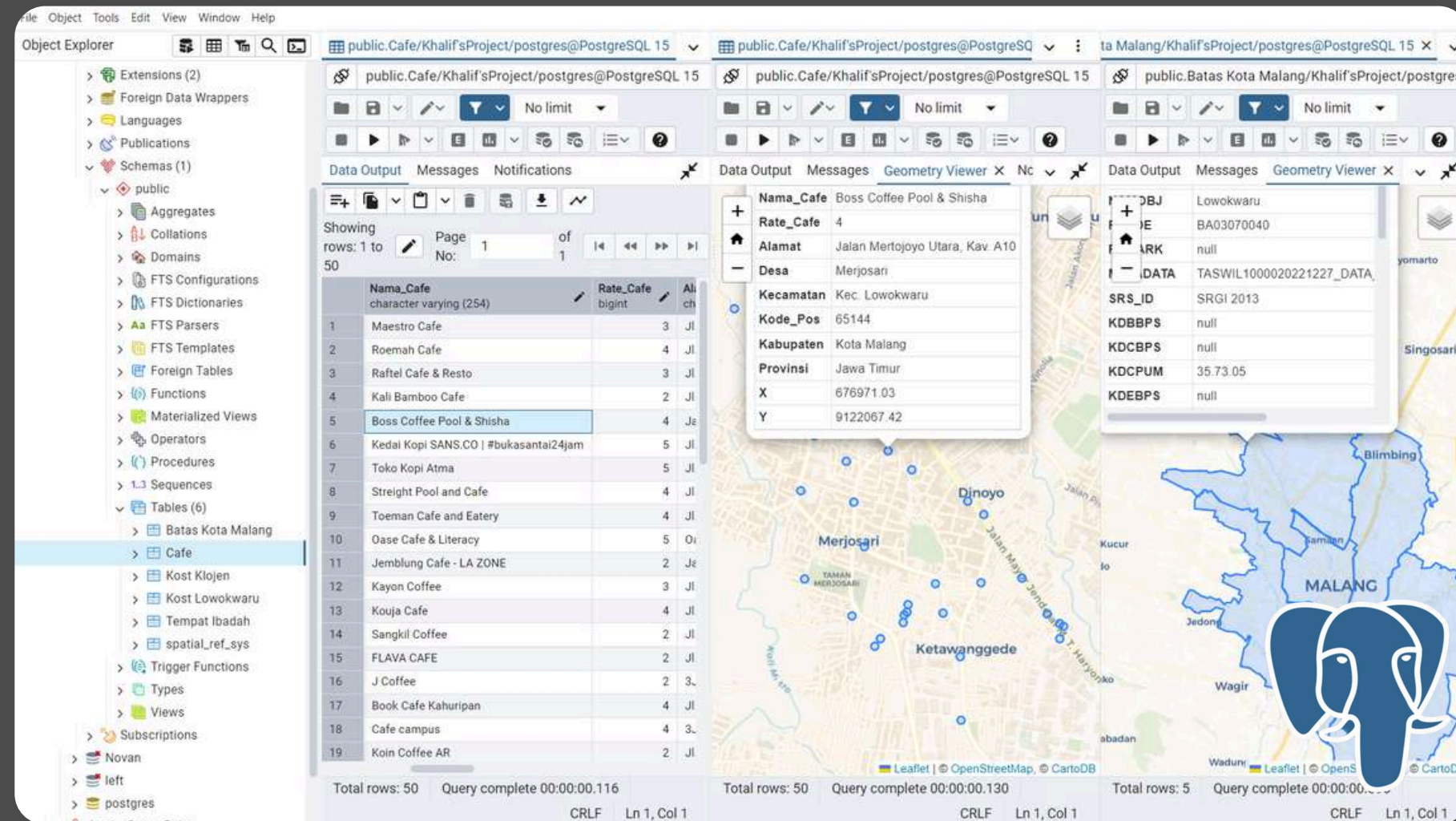
This project focuses on leveraging remote sensing and machine learning to analyze vegetation patterns using high-resolution imagery. By applying the Convolutional Neural Networks (CNN) method, the project aims to classify and spatially map vegetation areas. The development was carried out using the Python programming language with support from libraries such as rasterio, matplotlib, numpy, pandas, scikit-image, and Pillow to facilitate raster processing, spatial visualization, and computational image feature analysis.

	precision	recall	f1-score	support
0	0.96	0.89	0.92	938886
1	0.90	0.84	0.87	739332
2	0.82	0.95	0.88	1071984
3	0.98	0.87	0.92	395526
accuracy			0.89	3145728
macro avg	0.91	0.89	0.90	3145728
weighted avg	0.90	0.89	0.89	3145728



[Click here for More information about this project](#)

Web Map



Integration of WebMap with PostgreSQL and GeoServer as the spatial database and server

This project focuses on developing an interactive web-based mapping system for managing and visualizing spatial data efficiently. The system integrates PostgreSQL as the main database, enhanced with the PostGIS extension to store and analyze geospatial data such as points, lines, and polygons. GeoServer is used to publish spatial data online through standard web services (WMS/WFS), allowing users to access, visualize, and interact with maps directly from the web without the need for specialized GIS software.

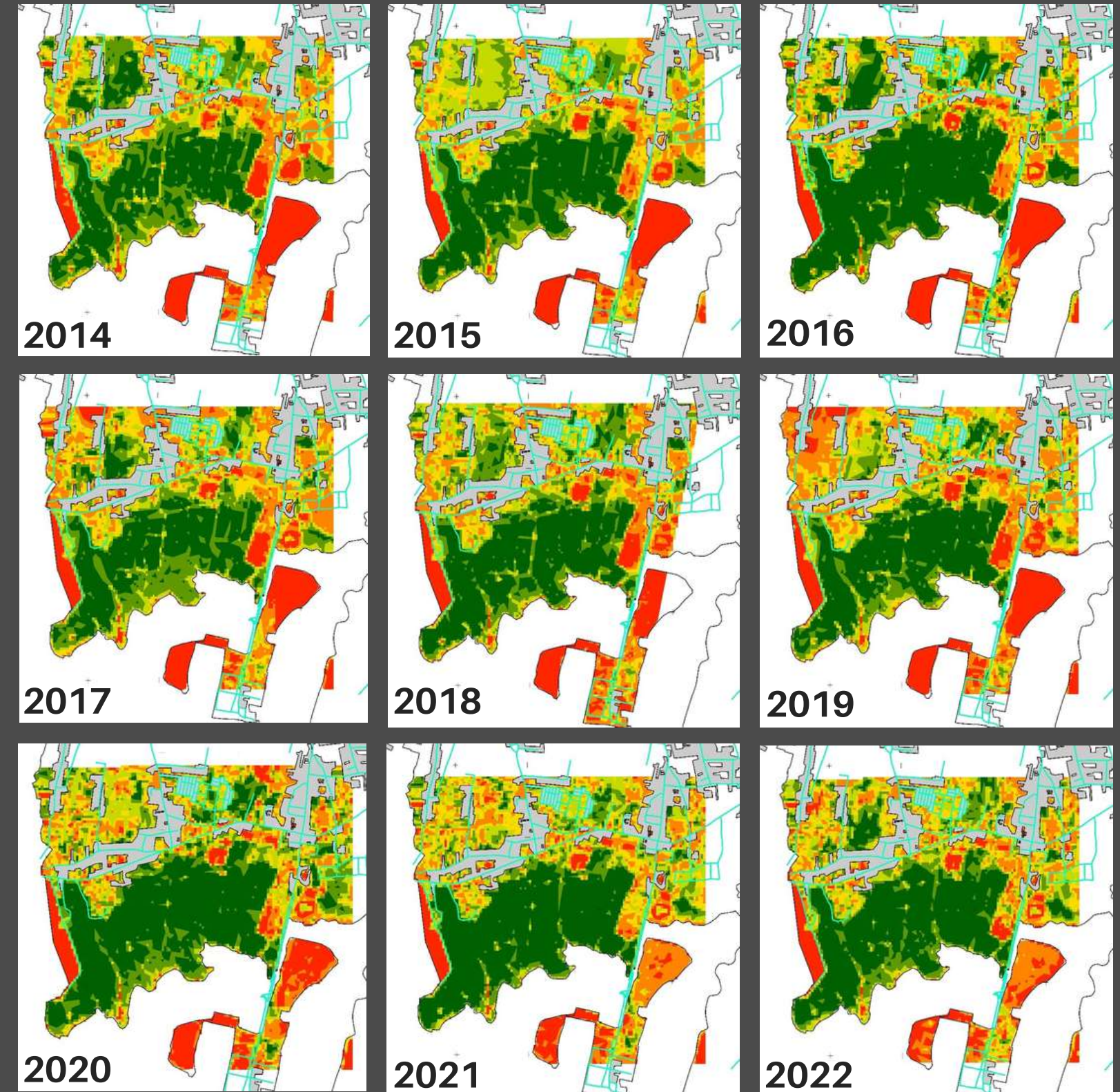
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Mangrove Changes



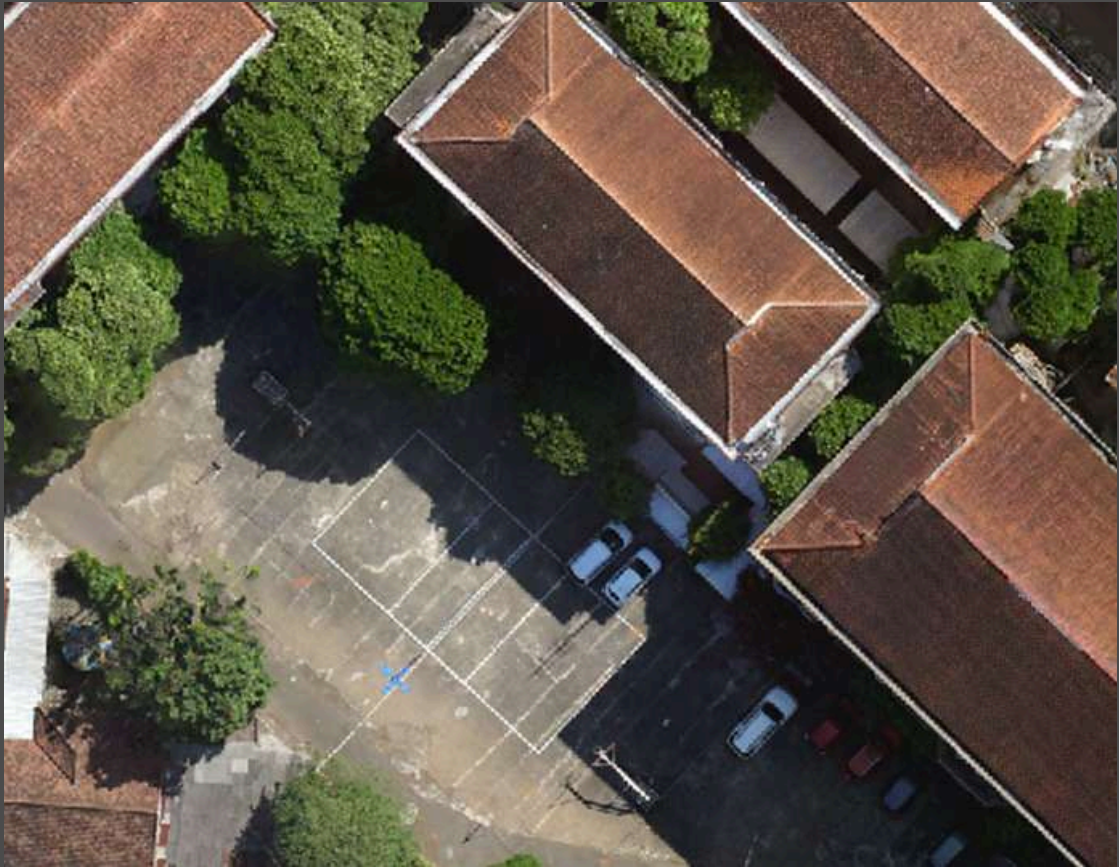
Mangrove Analysis using Google Earth Engine (Case Study: Pemogan, South Denpasar, Bali)

In this project, I analyzed mangrove vegetation changes from 2014 to 2023 using Google Earth Engine (GEE) and supporting GIS software. I started by collecting and processing multi-temporal satellite imagery (Landsat and Sentinel) through GEE, applying cloud masking, and calculating vegetation indices such as NDVI to measure mangrove density. The results were then exported, refined, and classified in ArcMap, including reclassification, raster-to-polygon conversion, and area calculation for each vegetation density class. To ensure accuracy, I compared the results between Landsat and Sentinel imagery and validated anomalies caused by cloud cover using alternative datasets from USGS Earth Explorer.



[Click here for More information about this project](#)

Orthophoto



Aerial Mapping

This project was carried out by a group in which I was responsible for data processing. It focuses on orthophoto generation using aerial imagery acquired from UAVs. The workflow involved the placement of Ground Control Points (GCPs) and Independent Control Points (ICPs) to ensure spatial accuracy. The process included photo alignment, dense cloud generation, mesh construction, and the creation of Digital Elevation Models (DEM) and orthomosaics for large-scale mapping. The final outputs achieved a ground resolution of 2.09 cm/pixel, with RMSE accuracy below 1 meter for both GCP and ICP validation.

**AERIAL MAP
OF THE SUMBERSARI VILLAGE AREA
LOWOKWARU DISTRICT, MALANG CITY,
EAST JAVA, INDONESIA**

Count	X error (m)	Y error (m)	Z error (m)	XY error (m)	Total (m)
4	0.113385	0.0392823	0.014451	0.119997	0.120864

Table 3. Control points RMSE.
X - Longitude, Y - Latitude, Z - Altitude.

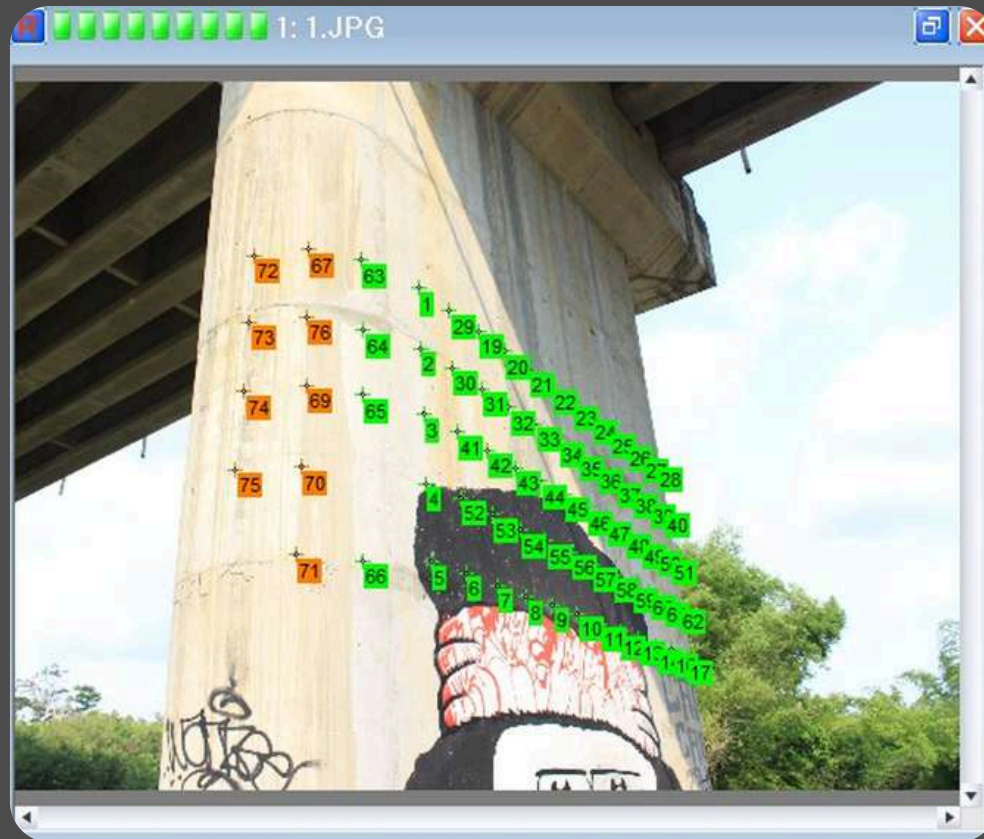
Count	X error (m)	Y error (m)	Z error (m)	XY error (m)	Total (m)
5	0.208833	0.277197	0.938687	0.347059	1.00079

Table 4. Check points RMSE.



[Click here for More information about this project](#)

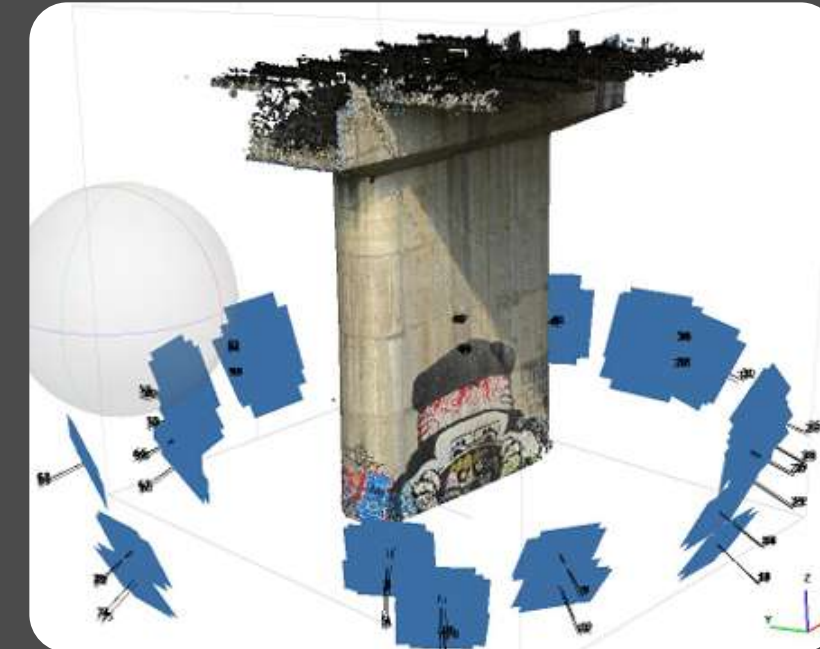
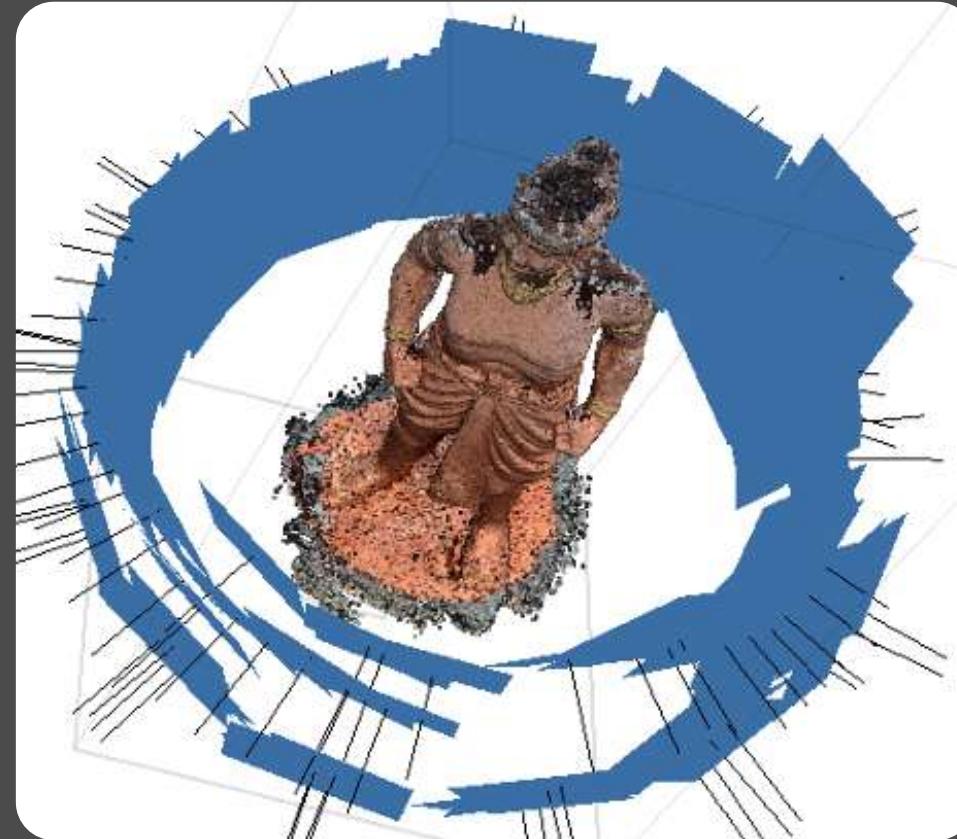
3D Modelling



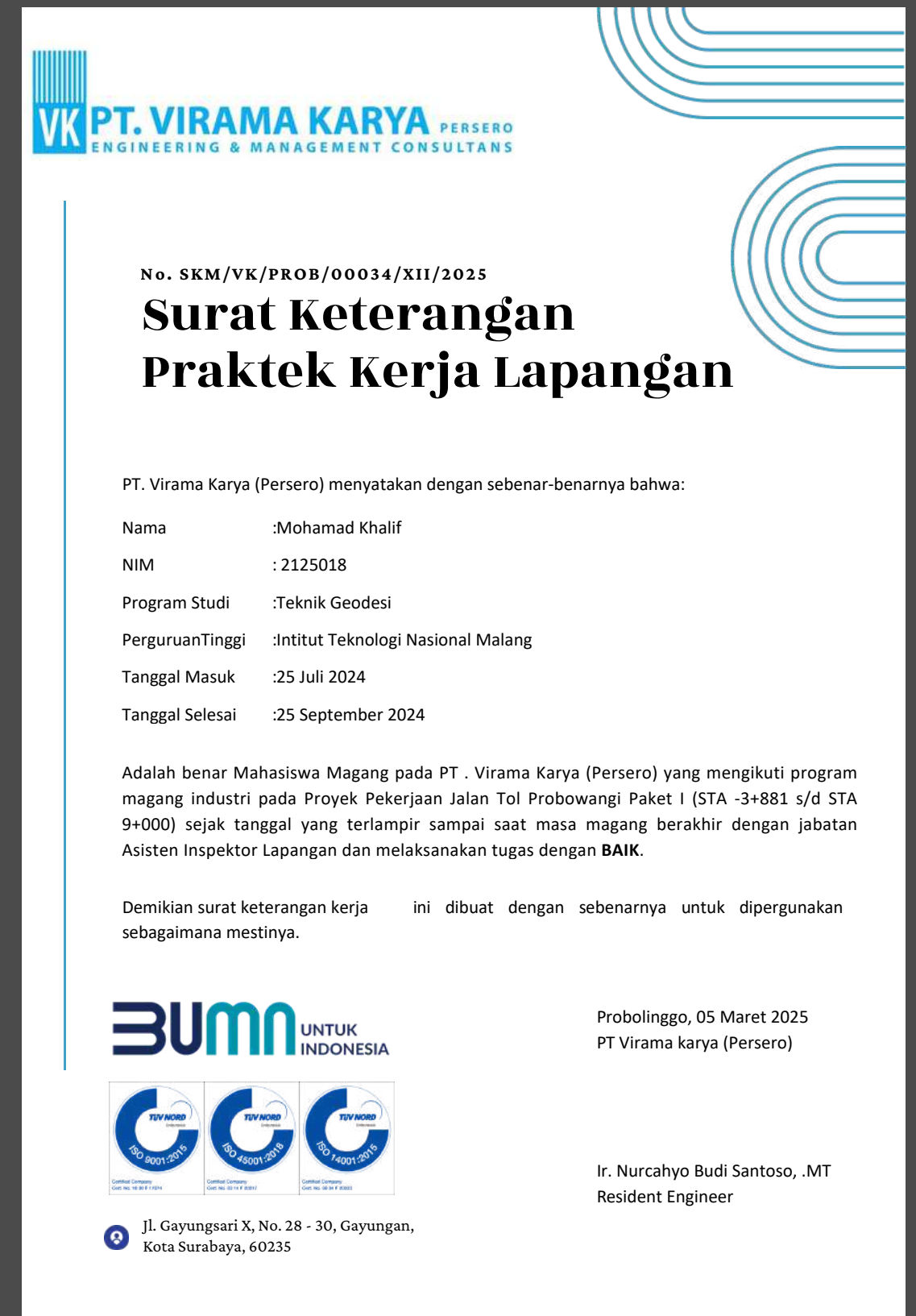
3D Model

This 3D modeling project utilized a smartphone camera with convergent image acquisition and was processed using Agisoft Metashape. The main purpose of this project is to monitor and analyze structural deformation over time, focusing on detecting potential damage, measuring positional or shape changes, and ensuring the safety and integrity of infrastructures such as bridges and monuments.

[Click here for More information about this project](#)



Certificate of Internship



Certificate of Participation



TOEFL Prediction

